

Semantic Caching

The production guide: how it works, picking a threshold, what never to cache, and how to invalidate.

*"Close enough" is the whole trap.
Get the threshold right, or don't ship it.*

1. How it works

A normal cache matches exact text, so "refund policy?" and "how do refunds work?" are different keys and both hit the LLM.

- ☐ Embed the incoming question.
- ☐ Search the cache for the nearest past question.
- ☐ Similarity above your threshold? Return that cached answer. No LLM call.
- ☐ Below threshold? Call the LLM, then store the new (embedding, answer, metadata).

2. Worth it when

- ☐ Questions repeat a lot.
Support bots, FAQs, docs Q&A, common data questions.
- ☐ Answers are the same for everyone asking.
- ☐ Skip it for: personalized answers, a user's own data, real-time info, agent actions with side effects, or low-repetition workloads.

3. Picking a threshold

- ☐ Start strict, around 0.95 similarity.
A common starting range is 0.90-0.95.
- ☐ Build a test set from real traffic.
Same-meaning pairs, plus look-alike pairs that need different answers.
- ☐ Lower the threshold step by step and track the false-hit rate.
A cached answer served when it shouldn't be.
- ☐ Use different thresholds per query type.
One team's tuned values: FAQ 0.94, support 0.92, transactional 0.97.
- ☐ Tools use different scales.
RedisVL: cosine distance 0-2, lower = stricter, default 0.1. Not similarity.

4. What goes in the cache key

- ☐ Tenant / customer.
Never share cached answers across tenants.
- ☐ Language / locale.
- ☐ Model + system prompt version.
- ☐ Knowledge base / index version.

5. Invalidation

- ☐ Set a TTL on every entry.
- ☐ Bump the version when your prompt, model, or source docs change.
- ☐ Flush entries tied to any doc that was updated.

6. Classic false hits

- ☐ Different object.
“cancel my subscription” vs “cancel my order” → 0.87 similar in one team’s tests.
- ☐ Different entity.
“refund policy India” vs “refund policy US”
- ☐ Negation or different numbers.
“can I...” vs “why can’t I...”; different error codes or dates.

7. What to monitor

- ☐ Hit rate and latency saved per hit.
- ☐ False-hit rate.
Sample and review actual cache hits every week.
- ☐ LLM spend saved.

8. Tools

- ☐ RedisVL SemanticCache, GPTCache, a gateway-level cache, or DIY with pgvector.
GPTCache no longer adds new model adapters; use its generic get/set API.

*Not the same as prompt caching –
that still calls the model. This skips it.*

Sources: RedisVL docs (v0.26), GPTCache README, and a VentureBeat engineering write-up on semantic caching (Jan 2026). Checked Sep 28, 2026. @gouravkargwal